

Supplementary Information

The choice of stability reference changes what crystal-discovery benchmarks count as success

Supplementary Note 1 | Frozen support and physical endpoint layer

D2 is the 36,802-row cohort with a single unambiguous tolerance-based structural match across MP, MatterGen alex-mp-20 and official Alexandria-PBE. D5 intersects D2 with the four archived model-energy tables. D5 contains 36,801 rows, of which 120 are elemental targets. The remaining 36,681 compounds form the candidate and transductive reference batch. Four rankings and five physical coordinates are complete for 36,650 compounds, defining M_{phys} (Table S3). Candidate, reference-pool and evaluation-support manifests are frozen separately.

The physical layer has three source coordinates and two matched-pool coordinates. MP and Alexandria-PBE formation energies are reconstructed separately on the same matched D2 subsystem inventory. Matched-pool agreement, source consensus and the conservative audit assignment are derived outputs outside the five-coordinate model comparison.

Supplementary Note 2 | Withdrawn ranking construction and adopted estimand

A self-included predicted-hull construction places every structure that supports the predicted hull at zero distance. The four archived rankings produced zero-distance tie blocks of 15,688–19,190 structures, and every top-1000 boundary fell inside one of those blocks (Table S4). Row-identifier ordering consequently affected stable hits. The MACE-MP MP-native top-1000 result was 802 hits under row-identifier ordering; the analytic tie expectation is 706.23 with a 95% hypergeometric interval of 679–733 (Table S5). The associated self-included discovery curves, fixed top- K binary metrics and winner claims are withdrawn. The revised discovery curves are independently recomputed from the adopted equivalence-class-excluded score and use analytic expectations within every score tie.

The adopted score is a leave-one-tolerance-equivalence-class-out signed reference-hull margin. The candidate's complete connected equivalence component is removed from its compatible subsystem. The remaining D5 batch phases and elemental anchors form the predicted reference hull. Other candidates may enter that hull, making the score retrospective, batch-relative and transductive. Across 146,724 primary candidate–model evaluations, the complete exclusion class is absent from the reference pool and decomposition simplex. The largest primary score tie is four, and every top-1000 boundary tie has size one.

The archived raw energies per atom remain excluded from primary ranking because their MP-native AUROCs are 0.465–0.489; reversing direction gives 0.511–0.535 (Table S6). The previous self-included predicted-hull column in the historical audit file is superseded and intentionally omitted.

Supplementary Note 3 | Matching sensitivity

Cross-source matching uses reduced-formula prefiltering followed by StructureMatcher. The default parameters are $\text{ltol}=0.2$, $\text{stol}=0.3$ and $\text{angle_tol}=5^\circ$, with primitive reduction, scaling and supercell attempts enabled and subset matching disabled. Tight parameters are $(0.1, 0.2, 3^\circ)$ and loose parameters are $(0.3, 0.4, 7^\circ)$.

The cross-source sensitivity is a survival audit of the frozen D1 and D2 mappings. It does not search for new loose-tolerance matches. Tight matching retains 42,799 D1 rows and 35,877 D2 rows; default and loose matching retain all 43,139 D1 and 36,802 D2 rows (Table S7). The tight losses are distributed across many chemical systems; the largest individual systems contribute 11 rows each (Table S8).

The D5 equivalence graph is rebuilt across all three settings. Tight, default and loose matching yield 35,746, 35,740 and 35,449 components, respectively. Non-singleton components increase from 918 to 1,155, while the largest class remains six. All four rankings are recomputed for every setting. The default scores reproduce the primary ranking exactly. No setting produces a top-1000 boundary tie. Matching tolerance can change a close point ordering: under loose matching, M3GNet is the top-1000 point winner for all five physical coordinates, with first–second margins of 1–11 hits.

Supplementary Note 4 | Threshold and indeterminate-zone estimands

The official/native endpoint uses each source's released binary field. The source-coordinate threshold scan regenerates labels as $y_s(t) = \mathbf{1}[E_{\text{hull},s} \leq t]$. The $t = 0$ coordinate labels reproduce all available official labels. Thresholds of 10, 25 and 50 meV atom⁻¹ ask how a screening tolerance changes the same source-coordinate comparison (Table S9).

The indeterminate analysis keeps the base threshold at zero and changes the width w . Stable rows satisfy $E_{\text{hull}} \leq 0$, indeterminate rows satisfy $0 < E_{\text{hull}} < w$, and unstable rows satisfy $E_{\text{hull}} \geq w$. Robust conflicts require definite stable and definite unstable assignments across a source pair. Full-support and decisive-support rates use different denominators and are reported together (Table S10).

Supplementary Note 5 | Matched-pool operational categories

All native MP–Alexandria-PBE conflicts, reconstructable native conflicts and unreconstructable conflicts are separate columns. Reconstructable native conflicts equal phase-pool-sensitive plus persistent rows. Matched-pool conflicts equal persistent plus hidden rows. Table S11 retains these identities at every threshold. The categories describe outcome changes after phase-inventory alignment; they do not assign a unique computational or physical mechanism.

Supplementary Note 6 | Metrics, budgets and model decisions

Full-ranking evaluation reports AUROC, average precision (AP), prevalence and normalised AP, $nAP = (AP - \pi)/(1 - \pi)$ (Table S12). Stable hits, yield and excess hits are reported at fixed budgets. When a boundary intersects a tie, the point estimate is the analytic hypergeometric expectation; the tie-only exact interval and best–worst bounds remain separate. The primary top-1000 table is shown in Table S13.

Budget sensitivity is evaluated at $K = 100, 300, 500, 1,000$ and $5,000$. Table S14 reports point winners, first–second margins, bootstrap winner frequencies, conservative MP-selection regret and boundary tie occupancy. MACE-MP remains the winner for all three full-ranking metrics and all five physical coordinates in every bootstrap replicate (Table S15). At $K = 1000$, conservative median MP-selection regret is 0–3 hits across all five coordinates, or 1–3 hits across the four non-MP coordinates; 95% upper limits are 7–18 hits outside the MP anchor (Table S16).

Supplementary Note 7 | Bootstrap scope

Chemical systems are sampled with replacement 1,000 times using seed 20260719. Every selected cluster retains all rows, models and coordinates. Tie expectations, metrics, winners, margins and regret are recomputed within each replicate. The intervals describe cohort variation conditional on fixed model predictions, fixed D5 reference pool and fixed structural-equivalence classes. They do not quantify model-energy error, DFT-workflow uncertainty or reference-pool uncertainty.

The alternative-unit audit repeats the same calculations with reduced-formula and structural-equivalence-family units using seed 20260826. Switch-rate intervals use paired endpoint labels within each D2 replicate; model metrics and decision quantities use paired endpoints and models within each M_{phys} replicate. The available structural-equivalence family is used as the related-structure unit. Prototype and element-family supercluster labels are not available. Table S18 gives all switch-rate intervals, and Table S20 gives the complete zero-threshold $K = 1000$ winner, margin and regret summary. MACE-MP has frequency 1.00 for every full-ranking metric, endpoint and resampling unit (Table S19).

Complete CSV summaries for all four thresholds and all four budgets are included in the alternative-unit bootstrap source-data directory of the release package.

Supplementary Note 8 | Model and candidate provenance

Generative approaches construct candidate crystals using learned continuous representations,¹ variational and equivariant diffusion models,^{2,3} and property-conditioned generation.⁴

ALIGNN-FF, CHGNet, M3GNet and MACE-MP enter the primary comparison because all four archived energy tables cover D5. Their checkpoints and published training resources are listed in Table S17. Materials Project-derived trajectories contribute to CHGNet and MACE-MP training; the comparison is a retrospective benchmark audit and is not presented as out-of-distribution generalisation.

SourceAware endpoint assignments require one unambiguous tolerance-based structural match. Formula-only overlap reports chemistry support. Most MatterGen and PGCGM candidates have no structural match, so these outputs quantify matching coverage instead of source-specific stability success.

Among the 4,291 structurally matched candidates in the public CHGNet top-5,000 screening cohort, MP-native stable yield is 0.406, audit-policy stable yield is 0.260 and the source-uncertain fraction is 0.334. The MatterGen 5,000-candidate public formula cohort contains 421 formula-only overlaps and 4,579 unmatched rows. The PGCGM 3,000-candidate generated-crystal cohort contains 39 formula-only overlaps and 2,961 unmatched rows. Neither generated cohort has a tolerance-based structural match in this analysis.

Supplementary Note 9 | Source-energy workflow background

A source energy inherits the electronic-structure approximation and numerical settings used to calculate it. Relevant methodological foundations include the PBE generalized-gradient approximation,⁵ the projector augmented-wave method and its plane-wave implementation,^{6,7} and iterative algorithms for plane-wave total-energy calculations.⁸ The AFLOW standard documents how such settings are specified for high-throughput calculations.⁹ Cross-code comparisons of elemental equations of state demonstrate that numerical reproducibility can be assessed under controlled computational choices.¹⁰

Energy corrections address additional sources of error. Work on transition-metal oxides examines GGA oxidation-energy errors and Hubbard- U corrections;¹¹ GGA/GGA+ U mixing and fitted elemental-phase reference energies provide different approaches to improving formation enthalpies.^{12,13} Reaction-energy comparisons for ternary oxides connect the accuracy of relative energies to phase-stability assessment.¹⁴ These studies motivate retaining each source's energy convention when rebuilding a hull. Our reconstruction uses the released source energies and changes the competing-phase inventory.

Supplementary Note 10 | Predictive model background

Materials prediction methods use both composition and structure. Composition-based descriptors support general property prediction,¹⁵ and electron-configuration ensembles have been developed for stability classification.¹⁶ CGCNN and

MEGNet learn properties from crystal graphs,^{17,18} while SchNet uses continuous-filter convolutions and ALIGNN explicitly incorporates bond angles through a line graph.^{19,20} NequIP and Allegro develop equivariant representations for interatomic potentials.^{21,22}

The present evaluation uses archived predictions from ALIGNN-FF, CHGNet, M3GNet and MACE-MP. CHGNet, M3GNet and MACE foundation-model publications describe the corresponding model families.^{23–25} The ALIGNN citation above describes its architectural basis; the exact force-field checkpoint and all four training-resource records are given in Supplementary Table S17.

Training-data coverage and transfer are active areas of model development. Systematic softening has been reported for M3GNet, CHGNet and MACE-MP-0 in out-of-distribution atomistic tasks.²⁶ MP-ALOE expands off-equilibrium training data at the r^2 SCAN level,²⁷ SevenNet-Omni addresses transfer across heterogeneous databases,²⁸ and GRACE provides a further family of foundational interatomic potentials.²⁹ These developments provide context for future model comparisons; the present label-source comparison keeps its four archived predictions fixed.

Supplementary Note 11 | Predictive uncertainty and target definition

Stability-label provenance also matters when reporting predictive uncertainty. Monte Carlo dropout and deep ensembles estimate model uncertainty,^{30,31} while calibration methods assess the correspondence between confidence and observed correctness.³² Conformal prediction constructs prediction sets or intervals with coverage guarantees under assumptions such as exchangeability; conformalized quantile regression adapts intervals to variation across inputs.^{33–35} Selective classification trades prediction coverage for error control, and set-valued classification allows multiple plausible labels.^{36,37} Applying these tools to crystal stability requires declaring which database-defined outcome is being predicted. Our comparison supplies evidence about the sensitivity of that outcome to its reference.

Supplementary tables

Table S1: Source fields used to construct the physical endpoint layer. The listed native fields are continuous energy-above-hull coordinates. Binary labels are regenerated at the declared threshold and audited against the released native labels. Formation-energy inputs enter only the MP and Alexandria-PBE matched-pool coordinates.

Source	Snapshot / query	Native continuous field	Formation-energy field	Unit; threshold
Materials Project	MP Summary API, 2026-05-20	<code>energy_above_hull</code>	<code>formation_energy_per_atom</code>	eV atom ⁻¹ ; $\leq 10^{-8}$
MatterGen	MatterGen release archive;	<code>energy_above_hull</code>	unavailable in released score table	eV atom ⁻¹ ; $\leq 10^{-8}$
alex-mp-20	training and validation tables			
official	PBE 3D 2025.07.02; retrieved	<code>entries[].data.e_above_hull</code>	<code>entries[].data.e_form</code>	eV atom ⁻¹ ; $\leq 10^{-8}$
Alexandria-PBE	2026-07-05			

Table S2: Retained/excluded audit for the fixed physical-endpoint cohort. Candidate exclusions are the 31 compound rows without all five physical coordinates; they differ in coordinate availability and composition. Stable fractions use available coordinates, with the evaluable row count shown in parentheses for matched-pool coordinates. Reference-hull phase count is measured after target-equivalence-class exclusion.

Feature	M_{phys} retained $n = 36,650$	Candidate excluded $n = 31$
MP source-coordinate stable fraction	40.9%	22.6%
alex-mp-20 source-coordinate stable fraction	36.2%	25.8%
Alexandria-PBE source-coordinate stable fraction	33.1%	29.0%
MP matched-pool stable fraction	48.7% (36,650)	– (0)
Alexandria-PBE matched-pool stable fraction	54.5% (36,650)	35.5% (31)
Formula element count	3.03	2.97
Formula atom count	7.25	6.65
Structure site count	10.59	12.10
Reference-hull phase count after target exclusion	33.33	47.10
Unique chemical-system count	19294.00	23.00
Chemical-system coverage of candidate universe	100.0%	0.1%
Transition-metal-containing	75.2%	61.3%
Lanthanide-containing	36.4%	38.7%
Oxygen-containing	25.3%	51.6%
Halogen-containing	13.5%	9.7%

Table S3: Frozen cohort construction for the revised physical-endpoint evaluation.

Stage	Rows
D5 frozen four-score intersection	36,801
D5 compound candidate universe	36,681
five physical hull coordinates complete	36,650
four batch-relative rankings complete	36,681
Mphys fixed physical evaluation support	36,650

Table S4: Failure audit for the withdrawn self-included hull construction at $K = 1000$. Every boundary lies inside the zero-distance tie block.

Model	Boundary tie	Strictly before	Largest tie block
ALIGNN-FF	15,688	0	15,688
CHGNet	19,190	0	19,190
M3GNet	17,178	0	17,178
MACE-MP	17,364	0	17,364

Table S5: Illustrative consequence of row-identifier ordering in the withdrawn MACE-MP MP-native top-1000 analysis.

Row-ID hits	Tie expectation	95% low	95% high	Worst	Best
802	706.2	679	733	0	1,000

Table S6: Construct-validity audit of archived energy-per-atom scores. The withdrawn self-included predicted-hull column from the earlier output is intentionally omitted.

Model	Rows	Raw AUROC	Reverse AUROC
ALIGNN-FF	31,872	0.4655	0.5345
CHGNet	31,872	0.4681	0.5319
M3GNet	31,872	0.4809	0.5191
MACE-MP	31,872	0.4891	0.5109

Table S7: Tight/default/loose matching sensitivity. Cross-source counts measure survival of frozen mappings; each D5 equivalence graph and all model rankings are rebuilt.

Tolerance	D1	D2	M_{phys}	Classes	Non-singleton	Largest	Default score max diff
tight	42,799	35,877	35,745	35,746	918	6	–
default	43,139	36,802	36,650	35,740	922	6	0.0000
loose	43,139	36,802	36,650	35,449	1,155	6	–

Table S8: Chemical systems with the largest loss under tight matching relative to default matching.

Tolerance	Direction	Chemical system	Rows	All changed
tight	lost_vs_default	F-O-V	11	925
tight	lost_vs_default	Ga-Mg	11	925
tight	lost_vs_default	Li-Mg	11	925
tight	lost_vs_default	F-Li-V	8	925
tight	lost_vs_default	Ca-Mg	7	925
tight	lost_vs_default	Mo-S-Te-W	7	925
tight	lost_vs_default	Li-Mn-O	7	925
tight	lost_vs_default	F-Li-O-V	7	925
tight	lost_vs_default	F-Li-Mn	6	925
tight	lost_vs_default	Cu-F-O	6	925
tight	lost_vs_default	F-Li-Mn-O	5	925
tight	lost_vs_default	K-N-O	5	925
tight	lost_vs_default	H-Li-N	5	925
tight	lost_vs_default	Cu-F-Li	5	925
tight	lost_vs_default	O-Si	5	925

Table S9: Native-coordinate threshold scan on the full D2 support.

Threshold (meV atom ⁻¹)	Pair	Rows	Switches	Switch rate (%)
0	mp–alexmp20	36,802	4,244	11.53
0	mp–alex_pbe	36,802	5,666	15.40
0	alexmp20–alex_pbe	36,802	3,862	10.49
10	mp–alexmp20	36,802	3,923	10.66
10	mp–alex_pbe	36,802	4,886	13.28
10	alexmp20–alex_pbe	36,802	3,315	9.01
25	mp–alexmp20	36,802	3,258	8.85
25	mp–alex_pbe	36,802	4,173	11.34
25	alexmp20–alex_pbe	36,802	2,995	8.14
50	mp–alexmp20	36,802	2,396	6.51
50	mp–alex_pbe	36,802	3,291	8.94
50	alexmp20–alex_pbe	36,802	2,425	6.59

Table S10: MP–Alexandria-PBE indeterminate-zone analysis on reconstructable D2 rows.

Width (meV atom ⁻¹)	All rows	Decisive rows	Robust conflicts	All-support (%)	Decisive-support (%)
10	36,770	29,095	2,484	6.76	8.54
20	36,770	24,790	1,455	3.96	5.87
25	36,770	23,160	1,163	3.16	5.02
30	36,770	21,773	954	2.59	4.38
50	36,770	17,505	512	1.39	2.92

Table S11: Threshold-dependent MP–Alexandria-PBE operational conflict components.

Threshold	All native	Reconstructable	Incomplete	Phase-pool-sensitive	Persistent	Hidden	Matched-pool
0	5,666	5,661	5	3,659	2,002	2,895	4,897
10	4,886	4,881	5	2,946	1,935	2,858	4,793
25	4,173	4,169	4	2,343	1,826	2,485	4,311
50	3,291	3,285	6	1,612	1,673	1,748	3,421

Table S12: Full-ranking metrics at the official zero-threshold physical endpoints. AP and prevalence are reported with normalised AP.

Model	Endpoint	Rows	Positive	Prevalence	AUROC	AP	nAP
ALIGNN-FF	mp_source	36,650	15,001	0.4093	0.6987	0.6037	0.3291
ALIGNN-FF	alexmp20_source	36,650	13,266	0.3620	0.6836	0.5321	0.2666
ALIGNN-FF	alex_pbe_source	36,650	12,148	0.3315	0.6805	0.4893	0.2361
ALIGNN-FF	mp_matched_pool	36,650	17,844	0.4869	0.7263	0.7321	0.4779
ALIGNN-FF	alex_pbe_matched_pool	36,650	19,985	0.5453	0.7374	0.7880	0.5339
CHGNet	mp_source	36,650	15,001	0.4093	0.8313	0.7208	0.5273
CHGNet	alexmp20_source	36,650	13,266	0.3620	0.8121	0.6450	0.4435
CHGNet	alex_pbe_source	36,650	12,148	0.3315	0.8024	0.5954	0.3948
CHGNet	mp_matched_pool	36,650	17,844	0.4869	0.8721	0.8635	0.7341
CHGNet	alex_pbe_matched_pool	36,650	19,985	0.5453	0.9107	0.9282	0.8421
M3GNet	mp_source	36,650	15,001	0.4093	0.8019	0.7035	0.4980
M3GNet	alexmp20_source	36,650	13,266	0.3620	0.7844	0.6299	0.4199
M3GNet	alex_pbe_source	36,650	12,148	0.3315	0.7767	0.5841	0.3779
M3GNet	mp_matched_pool	36,650	17,844	0.4869	0.8363	0.8362	0.6807
M3GNet	alex_pbe_matched_pool	36,650	19,985	0.5453	0.8679	0.8999	0.7798
MACE-MP	mp_source	36,650	15,001	0.4093	0.8673	0.7598	0.5933
MACE-MP	alexmp20_source	36,650	13,266	0.3620	0.8490	0.6871	0.5097
MACE-MP	alex_pbe_source	36,650	12,148	0.3315	0.8419	0.6403	0.4620
MACE-MP	mp_matched_pool	36,650	17,844	0.4869	0.8982	0.8885	0.7828
MACE-MP	alex_pbe_matched_pool	36,650	19,985	0.5453	0.9370	0.9530	0.8966

Table S13: Tie-aware stable hits and yield at $K = 1000$.

Model	Endpoint	Hits	Yield	Random hits	Excess hits	Boundary tie
ALIGNN-FF	mp_source	710.0	0.7100	409.3	300.7	1
ALIGNN-FF	alexmp20_source	615.0	0.6150	362.0	253.0	1
ALIGNN-FF	alex_pbe_source	554.0	0.5540	331.5	222.5	1
ALIGNN-FF	mp_matched_pool	929.0	0.9290	486.9	442.1	1
ALIGNN-FF	alex_pbe_matched_pool	973.0	0.9730	545.3	427.7	1
CHGNet	mp_source	729.0	0.7290	409.3	319.7	1
CHGNet	alexmp20_source	645.0	0.6450	362.0	283.0	1
CHGNet	alex_pbe_source	576.0	0.5760	331.5	244.5	1
CHGNet	mp_matched_pool	970.0	0.9700	486.9	483.1	1
CHGNet	alex_pbe_matched_pool	1,000.0	1.00	545.3	454.7	1
M3GNet	mp_source	754.0	0.7540	409.3	344.7	1
M3GNet	alexmp20_source	677.0	0.6770	362.0	315.0	1
M3GNet	alex_pbe_source	605.0	0.6050	331.5	273.5	1
M3GNet	mp_matched_pool	973.0	0.9730	486.9	486.1	1
M3GNet	alex_pbe_matched_pool	999.0	0.9990	545.3	453.7	1
MACE-MP	mp_source	759.0	0.7590	409.3	349.7	1
MACE-MP	alexmp20_source	667.0	0.6670	362.0	305.0	1
MACE-MP	alex_pbe_source	598.0	0.5980	331.5	266.5	1
MACE-MP	mp_matched_pool	970.0	0.9700	486.9	483.1	1
MACE-MP	alex_pbe_matched_pool	996.0	0.9960	545.3	450.7	1

Table S14: Budget-dependent point winners, margins and conservative MP-selection regret. Tied point winners occupy separate rows.

K	Endpoint	Point winner	Margin	Winner freq.	Regret median	Regret 95% high	Boundary tie
100	alex_pbe_matched_pool	ALIGNN-FF	0.0000	1.00	0.0000	0.0000	1
100	alex_pbe_matched_pool	CHGNet	0.0000	1.00	0.0000	0.0000	1
100	alex_pbe_matched_pool	M3GNet	0.0000	1.00	0.0000	0.0000	1
100	alex_pbe_matched_pool	MACE-MP	0.0000	1.00	0.0000	0.0000	1
100	alex_pbe_source	M3GNet	2.00	0.6790	1.00	7.00	1
100	alexmp20_source	ALIGNN-FF	0.0000	0.3530	1.00	6.00	1
100	alexmp20_source	M3GNet	0.0000	0.4210	1.00	6.00	1
100	alexmp20_source	MACE-MP	0.0000	0.4000	1.00	6.00	1
100	mp_matched_pool	ALIGNN-FF	0.0000	1.00	0.0000	0.0000	1
100	mp_matched_pool	CHGNet	0.0000	1.00	0.0000	0.0000	1
100	mp_matched_pool	M3GNet	0.0000	1.00	0.0000	0.0000	1
100	mp_matched_pool	MACE-MP	0.0000	1.00	0.0000	0.0000	1
100	mp_source	M3GNet	0.0000	0.5480	0.0000	0.0000	1
100	mp_source	MACE-MP	0.0000	0.3530	0.0000	0.0000	1
300	alex_pbe_matched_pool	ALIGNN-FF	0.0000	1.00	0.0000	0.0000	1
300	alex_pbe_matched_pool	CHGNet	0.0000	1.00	0.0000	0.0000	1
300	alex_pbe_matched_pool	M3GNet	0.0000	1.00	0.0000	0.0000	1
300	alex_pbe_matched_pool	MACE-MP	0.0000	1.00	0.0000	0.0000	1
300	alex_pbe_source	M3GNet	8.00	0.9030	0.0000	9.00	1
300	alexmp20_source	M3GNet	8.00	0.8640	0.0000	8.00	1
300	mp_matched_pool	M3GNet	1.00	0.3830	0.0000	3.00	1
300	mp_source	M3GNet	7.00	0.7830	0.0000	0.0000	1
500	alex_pbe_matched_pool	CHGNet	0.0000	1.00	0.0000	2.00	1
500	alex_pbe_matched_pool	M3GNet	0.0000	1.00	0.0000	2.00	1
500	alex_pbe_matched_pool	MACE-MP	0.0000	1.00	0.0000	2.00	1
500	alex_pbe_source	M3GNet	1.00	0.6080	0.0000	12.00	1
500	alexmp20_source	MACE-MP	1.00	0.3680	0.0000	9.00	1
500	mp_matched_pool	M3GNet	0.0000	0.7660	0.0000	6.02	1
500	mp_matched_pool	MACE-MP	0.0000	0.9040	0.0000	6.02	1
500	mp_source	MACE-MP	1.00	0.4330	0.0000	0.0000	1
1,000	alex_pbe_matched_pool	CHGNet	1.00	0.9330	3.00	7.00	1
1,000	alex_pbe_source	M3GNet	7.00	0.7640	1.00	17.00	1
1,000	alexmp20_source	M3GNet	10.00	0.8570	3.00	18.00	1
1,000	mp_matched_pool	M3GNet	3.00	0.7230	1.00	7.00	1
1,000	mp_source	MACE-MP	5.00	0.7770	0.0000	0.0000	1
5,000	alex_pbe_matched_pool	MACE-MP	19.00	0.9930	0.0000	0.0000	1
5,000	alex_pbe_source	MACE-MP	118.0	1.00	0.0000	0.0000	1
5,000	alexmp20_source	MACE-MP	129.0	1.00	0.0000	0.0000	1
5,000	mp_matched_pool	MACE-MP	12.00	0.8310	0.0000	13.00	1
5,000	mp_source	MACE-MP	85.00	1.00	0.0000	0.0000	1

Table S15: MACE-MP winner frequencies for full-ranking metrics at zero threshold.

Endpoint	Metric	Winning replicates	Frequency	Replicates
mp_source	auroc	1,000	1.00	1,000
mp_source	ap	1,000	1.00	1,000
mp_source	normalized_ap	1,000	1.00	1,000
alexmp20_source	auroc	1,000	1.00	1,000
alexmp20_source	ap	1,000	1.00	1,000
alexmp20_source	normalized_ap	1,000	1.00	1,000
alex_pbe_source	auroc	1,000	1.00	1,000
alex_pbe_source	ap	1,000	1.00	1,000
alex_pbe_source	normalized_ap	1,000	1.00	1,000
mp_matched_pool	auroc	1,000	1.00	1,000
mp_matched_pool	ap	1,000	1.00	1,000
mp_matched_pool	normalized_ap	1,000	1.00	1,000
alex_pbe_matched_pool	auroc	1,000	1.00	1,000
alex_pbe_matched_pool	ap	1,000	1.00	1,000
alex_pbe_matched_pool	normalized_ap	1,000	1.00	1,000

Table S16: Bootstrap MP-selection regret at $K = 1000$. Minimum and maximum columns retain MP-winner ties within a replicate.

Endpoint	Min median	Min 95% high	Max median	Max 95% high	Positive freq.
alex_pbe_matched_pool	3.00	7.00	3.00	7.00	0.8770
alex_pbe_source	0.0000	17.00	1.00	17.00	0.5200
alexmp20_source	2.00	17.00	3.00	18.00	0.5970
mp_matched_pool	1.00	7.00	1.00	7.00	0.5390
mp_source	0.0000	0.0000	0.0000	0.0000	0.0000

Table S17: Model checkpoint and published training-corpus provenance.

Model	Checkpoint	Published training corpus	Membership note
ALIGNN-FF	v12.2.2024_dft_3d_307k	JARVIS-DFT 3D	JARVIS-linked identifiers audited separately
CHGNet	CHGNet.load()	MPtrj	Materials Project-derived trajectories
M3GNet	M3GNet-PES-MatPES-PBE-2025.2	MatPES-PBE	Release-specific membership identifiers required
MACE-MP	mace-mp small	Materials Project trajectories	Materials Project-derived trajectories

Table S18: Alternative-unit bootstrap switch-rate intervals. Rates are percentages; intervals are the 2.5th and 97.5th percentiles across 1,000 replicates, and the median is shown separately.

Resampling unit	Threshold	Endpoint pair	95% low	Median	95% high
Chemical system	0	alex-mp-20 / Alexandria-PBE	10.14	10.50	10.86
Chemical system	0	MP / Alexandria-PBE	14.94	15.40	15.87
Chemical system	0	MP / alex-mp-20	11.14	11.52	11.93
Chemical system	10	alex-mp-20 / Alexandria-PBE	8.69	9.01	9.38
Chemical system	10	MP / Alexandria-PBE	12.84	13.27	13.70
Chemical system	10	MP / alex-mp-20	10.27	10.66	11.04
Chemical system	25	alex-mp-20 / Alexandria-PBE	7.79	8.14	8.48
Chemical system	25	MP / Alexandria-PBE	10.91	11.34	11.78
Chemical system	25	MP / alex-mp-20	8.51	8.86	9.25
Chemical system	50	alex-mp-20 / Alexandria-PBE	6.25	6.59	6.96
Chemical system	50	MP / Alexandria-PBE	8.53	8.93	9.36
Chemical system	50	MP / alex-mp-20	6.18	6.51	6.83
Equivalence family	0	alex-mp-20 / Alexandria-PBE	10.16	10.49	10.76
Equivalence family	0	MP / Alexandria-PBE	15.00	15.38	15.75
Equivalence family	0	MP / alex-mp-20	11.19	11.52	11.83
Equivalence family	10	alex-mp-20 / Alexandria-PBE	8.74	9.01	9.29
Equivalence family	10	MP / Alexandria-PBE	12.94	13.27	13.61
Equivalence family	10	MP / alex-mp-20	10.35	10.66	10.95
Equivalence family	25	alex-mp-20 / Alexandria-PBE	7.85	8.14	8.43
Equivalence family	25	MP / Alexandria-PBE	11.01	11.34	11.67
Equivalence family	25	MP / alex-mp-20	8.58	8.85	9.14
Equivalence family	50	alex-mp-20 / Alexandria-PBE	6.33	6.58	6.85
Equivalence family	50	MP / Alexandria-PBE	8.65	8.94	9.25
Equivalence family	50	MP / alex-mp-20	6.27	6.52	6.77
Reduced formula	0	alex-mp-20 / Alexandria-PBE	10.17	10.50	10.81
Reduced formula	0	MP / Alexandria-PBE	15.03	15.40	15.80
Reduced formula	0	MP / alex-mp-20	11.20	11.53	11.87
Reduced formula	10	alex-mp-20 / Alexandria-PBE	8.70	9.00	9.31
Reduced formula	10	MP / Alexandria-PBE	12.90	13.28	13.65
Reduced formula	10	MP / alex-mp-20	10.29	10.66	11.02
Reduced formula	25	alex-mp-20 / Alexandria-PBE	7.84	8.13	8.44
Reduced formula	25	MP / Alexandria-PBE	10.96	11.34	11.70
Reduced formula	25	MP / alex-mp-20	8.52	8.84	9.18
Reduced formula	50	alex-mp-20 / Alexandria-PBE	6.31	6.59	6.89
Reduced formula	50	MP / Alexandria-PBE	8.61	8.95	9.29
Reduced formula	50	MP / alex-mp-20	6.24	6.51	6.79

Table S19: Alternative-unit audit of full-ranking winners at zero threshold. MACE-MP won all 1,000 replicates for AUROC, AP and normalised AP at every physical endpoint and for every resampling unit.

Resampling unit	Endpoint	AUROC winner	AP winner	nAP winner	Replicates
Chemical system	MP source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Chemical system	alex-mp-20 source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Chemical system	Alexandria-PBE source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Chemical system	MP matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Chemical system	Alexandria-PBE matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Reduced formula	MP source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Reduced formula	alex-mp-20 source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Reduced formula	Alexandria-PBE source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Reduced formula	MP matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Reduced formula	Alexandria-PBE matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Equivalence family	MP source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Equivalence family	alex-mp-20 source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Equivalence family	Alexandria-PBE source	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Equivalence family	MP matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000
Equivalence family	Alexandria-PBE matched pool	MACE-MP (1.000)	MACE-MP (1.000)	MACE-MP (1.000)	1,000

Table S20: Alternative-unit audit of zero-threshold top-1000 decisions. Winner frequencies list every model with non-zero frequency. Margins and regrets report the median to 97.5th percentile range across 1,000 replicates; regret is the maximum regret among models tied as the MP-coordinate winner.

Resampling unit	Endpoint	Point-winner frequency	First-second margin	Max MP-selection regret
Chemical system	Alexandria-PBE matched pool	CHGNet 0.742; M3GNet 0.246; MACE-MP 0.012	2–7	3–7
Chemical system	Alexandria-PBE source	M3GNet 0.757; MACE-MP 0.242	7–23	2–18
Chemical system	alex-mp-20 source	M3GNet 0.846; MACE-MP 0.154	9–25	3–18
Chemical system	MP matched pool	CHGNet 0.108; M3GNet 0.653; MACE-MP 0.238	2–8	1–7
Chemical system	MP source	M3GNet 0.254; MACE-MP 0.747	7–27	–
Reduced formula	Alexandria-PBE matched pool	CHGNet 0.744; M3GNet 0.248; MACE-MP 0.007	2–8	3–8
Reduced formula	Alexandria-PBE source	M3GNet 0.765; MACE-MP 0.235	7–24	1–17
Reduced formula	alex-mp-20 source	M3GNet 0.843; MACE-MP 0.158	8–26	3–17
Reduced formula	MP matched pool	CHGNet 0.116; M3GNet 0.679; MACE-MP 0.206	2–8	1–7
Reduced formula	MP source	M3GNet 0.259; MACE-MP 0.742	7–26	–
Equivalence family	Alexandria-PBE matched pool	CHGNet 0.750; M3GNet 0.241; MACE-MP 0.009	2–7	3–8
Equivalence family	Alexandria-PBE source	M3GNet 0.737; MACE-MP 0.264	7–24	1–17
Equivalence family	alex-mp-20 source	M3GNet 0.844; MACE-MP 0.156	9–26	3–18
Equivalence family	MP matched pool	CHGNet 0.132; M3GNet 0.634; MACE-MP 0.234	2–8	1–7
Equivalence family	MP source	M3GNet 0.253; MACE-MP 0.747	6–27	–

Table S21: Derived agreement and selection rules. These combine stability assignments and remain outside the five-definition model comparison.

Rule	Inputs	Operation	Use
Matched-pool agreement	Both reconstructed coordinates	Status aggregation	Derived agreement
Consensus selection	Source and reconstruction statuses	Status aggregation	Derived selection
Audit policy	Source and reconstruction statuses	Status aggregation	Conservative policy

Table S22: Minimum SourceAware-Stability reporting checklist. These declarations make a stability endpoint and its model consequences auditable across future crystal-discovery benchmarks.

Reporting item	Minimum declaration
Source record	Database, version or query date, continuous native field, unit, formation-energy availability and relevant workflow metadata.
Matching and denominator	Tolerance-based matching rule, duplicate policy, denominator flow and retained/excluded cohort audit.
Physical endpoint layer	Native coordinates, matched-pool coordinates, phase inventory and separation from consensus or audit policies.
Decision boundary	Official threshold, analyst-defined threshold scan and indeterminate-zone convention.
Conflict burden	Pairwise switch rates, denominators and matched-pool operational components.
Ranking and validation	Frozen support, ranking estimand, tie policy, stable hits and yield at declared validation budgets.
Model decision	Point winner, winner frequency, first–second margin and cross-endpoint selection regret.
External scope and release	Matching coverage for generated or unmatched pools, frozen release, manifest and claim-to-output map.

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